Derivation of equation of convolution sum:

Let $x[n] = \{a, b, c, d, \ldots \ldots \ldots\}$

We can write $x[n]$ as:

$$x[n] = x[0]\,\delta[n-0] + x[1]\,\delta[n-1] + x[2]\,\delta[n-2] + \cdots$$

$$= a\,\delta[n] + b\,\delta[n-1] + c\,\delta[n-2] + \cdots \quad \cdots$$

$$\therefore x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

$\therefore$ Any discrete time signal can be represented as weighted sum of unit impulses.

Now, Let a system be:

$$x[n] \longrightarrow \underset{F\{\ \}}{\boxed{\text{System}}} \longrightarrow y[n] = F\{x[n]\}$$

$$= F\left\{\sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]\right\}$$

$$\therefore y[n] = T\{a\,\delta[n]\} + T\{b\,\delta[n-1]\} + T\{c\,\delta[n-2]\} + \cdots$$

[Linearity]

$$= \sum_{k=-\infty}^{\infty} T\{x[k]\,\delta[n-k]\}$$

Here $x[k]$ is independent of n and is a constant.

$$\therefore y[n] = \sum_{k=-\infty}^{\infty} x[k]\,T\{\delta[n-k]\} \quad \text{[Linearity]}$$

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Now in the given system, if $x[n]$ is input, output is $T\{x[n]\}$

$\therefore$ If $\delta[n]$ is input, output is $T\{\delta[n]\}$ and due to the property of time invariance, if $\delta[n-k]$ is input, output is $T\{\delta[n-k]\}$

We call the output of the system when its input is $\delta[n]$ as impulse response and is denoted by $h[n]$

$$\therefore T\{\delta[n-k]\} = h[n-k]$$

$$\therefore y[n] = \sum_{K=-\infty}^{\infty} x[K]\, h[n-k]$$

This is called convolution of input $x[n]$ with $\delta[n]$

$$\therefore y[n] = x[n] * \delta[n] = \sum_{K=-\infty}^{\infty} x[k]\, h[n-k]$$

$\therefore$ Generalized equation of convolution of $x_1[n]$ and $x_2[n]$ is

$$x_1[n] * x_2[n] = \sum_{K=-\infty}^{\infty} x_1[k]\, x_2[n-k]$$

(Ans)

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i) $x[n] = \{\underset{\uparrow}{1}, 0, 1\}$

$h[n] = \{\underset{\uparrow}{1}, 2, 1\}$

$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$$y[0] = \sum_{k=-\infty}^{\infty} x[k]\, h[k]$$

$$\begin{array}{r} 1\ 0\ 1 \\ 1\ 2\ 1 \\ \hline 0+0+1+0+0=1 \end{array}$$

$$= 1$$

$$y[1] = \sum_{k=-\infty}^{\infty} x[k]\, h[1-k]$$

$$\begin{array}{r} 1\ 0\ 1 \\ 1\ 2\ 1 \\ \hline 0+2+0+0=2 \end{array}$$

$$= 2$$

$$y[2] = \sum_{k=-\infty}^{\infty} x[k]\, h[2-k]$$

$$\begin{array}{r} 1\ 0\ 1 \\ 1\ 2\ 1 \\ \hline 1+0+1=2 \end{array}$$

$$= 2$$

$$y[3] = \sum_{k=-\infty}^{\infty} x[k]\, h[3-k]$$

$$\begin{array}{r} 1\ 0\ 1 \\ 1\ 2\ 1 \\ \hline 0+0+2+0=2 \end{array}$$

$$= 2$$

$$y[4] = \sum_{k=-\infty}^{\infty} x[k]\, h[4-k]$$

$$\begin{array}{r} 1\ 0\ 1 \\ 1\ 2\ 1 \\ \hline 1 \end{array}$$

$$= 1$$

$$\therefore y[n] = [\underset{\uparrow}{1}, 2, 2, 2, 1] \quad \text{(Ans)}$$